

PRACTICE ASSESSMENT 3

(SQL Joins - Dataset and Questions)

Sample Tables Setup

-- Create Database

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

-- Students Table

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    age INT,  
    department_id INT  
);
```

-- Courses Table

```
CREATE TABLE Courses (  
    course_id INT PRIMARY KEY,  
    course_name VARCHAR(100),  
    department_id INT,  
    instructor_id INT  
);
```

-- Enrollments Table (Many-to-Many)

```
CREATE TABLE Enrollments (  
    enrollment_id INT PRIMARY KEY,  
    student_id INT,  
    course_id INT,  
    grade VARCHAR(2)
```

```

);

-- Instructors Table

CREATE TABLE Instructors (
instructor_id INT PRIMARY KEY,
name VARCHAR(50),
department_id INT
);

-- Departments Table

CREATE TABLE Departments (
department_id INT PRIMARY KEY,
department_name VARCHAR(100)
);

-- Insert Sample Data

INSERT INTO Departments VALUES
(1, 'Computer Science'),
(2, 'Electrical Engineering'),
(3, 'Mathematics');

INSERT INTO Students VALUES
(101, 'Alice', 20, 1),
(102, 'Bob', 21, 2),
(103, 'Charlie', 22, 1),
(104, 'Diana', 23, 3);

INSERT INTO Instructors VALUES
(201, 'Dr. Smith', 1),
(202, 'Dr. Johnson', 2),
(203, 'Dr. Williams', 3);

INSERT INTO Courses VALUES
(301, 'Database Systems', 1, 201),

```

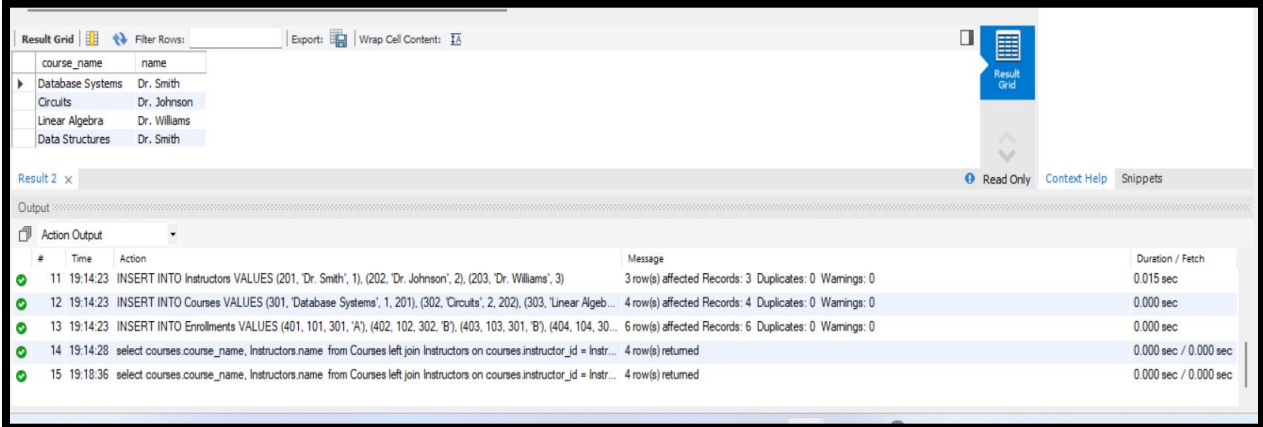
```
(302, 'Circuits', 2, 202),  
(303, 'Linear Algebra', 3, 203),  
(304, 'Data Structures', 1, 201);  
INSERT INTO Enrollments VALUES  
(401, 101, 301, 'A'),  
(402, 102, 302, 'B'),  
(403, 103, 301, 'B'),  
(404, 104, 303, 'A'),  
(405, 101, 304, 'A'),  
(406, 103, 304, 'C');
```

ASSESSMENT QUESTIONS:

Q1. List all courses with their instructor names.

```
select courses.course_name, Instructors.name
```

```
from Courses left join Instructors on courses.instructor_id = Instructors.instructor_id;
```



course_name	name
Database Systems	Dr. Smith
Circuits	Dr. Johnson
Linear Algebra	Dr. Williams
Data Structures	Dr. Smith

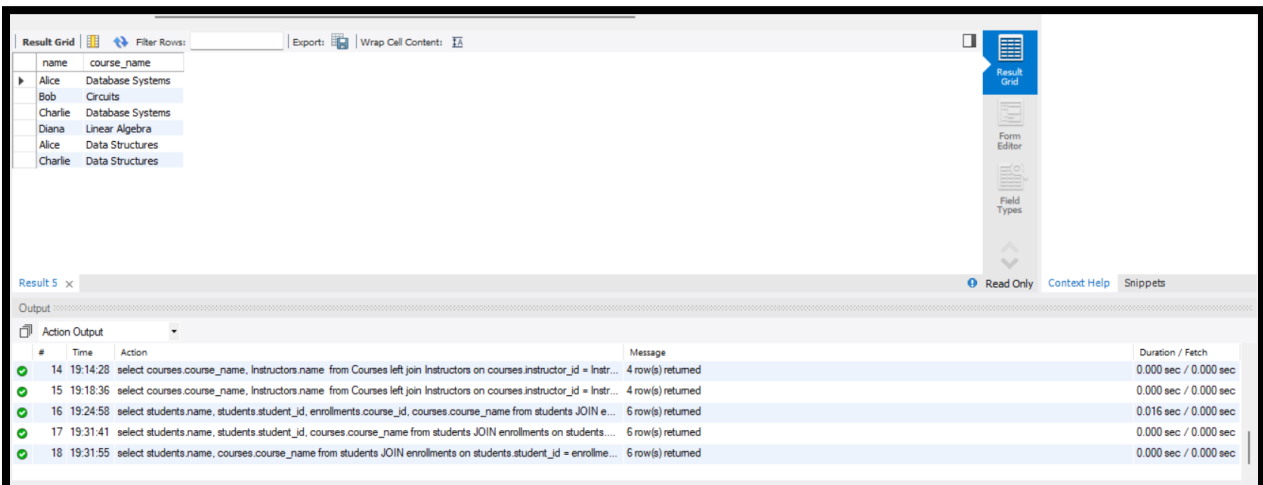
#	Time	Action	Message	Duration / Fetch
11	19:14:23	INSERT INTO Instructors VALUES (201, 'Dr. Smith', 1), (202, 'Dr. Johnson', 2), (203, 'Dr. Williams', 3)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
12	19:14:23	INSERT INTO Courses VALUES (301, 'Database Systems', 1, 201), (302, 'Circuits', 2, 202), (303, 'Linear Algebra', 3, 203), (304, 'Data Structures', 4, 204)	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
13	19:14:23	INSERT INTO Enrollments VALUES (401, 101, 301, 'A'), (402, 102, 302, 'B'), (403, 103, 301, 'B'), (404, 104, 303, 'A'), (405, 105, 304, 'A'), (406, 106, 304, 'B')	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0	0.000 sec
14	19:14:28	select courses.course_name, Instructors.name from Courses left join Instructors on courses.instructor_id = Instructors.instructor_id	4 row(s) returned	0.000 sec / 0.000 sec
15	19:18:36	select courses.course_name, Instructors.name from Courses left join Instructors on courses.instructor_id = Instructors.instructor_id	4 row(s) returned	0.000 sec / 0.000 sec

Q2. Show student names along with the course names they are enrolled in.

```
select students.name, courses.course_name
```

```
from students JOIN enrollments on students.student_id = enrollments.student_id
```

```
JOIN courses on enrollments.course_id = courses.course_id ;
```

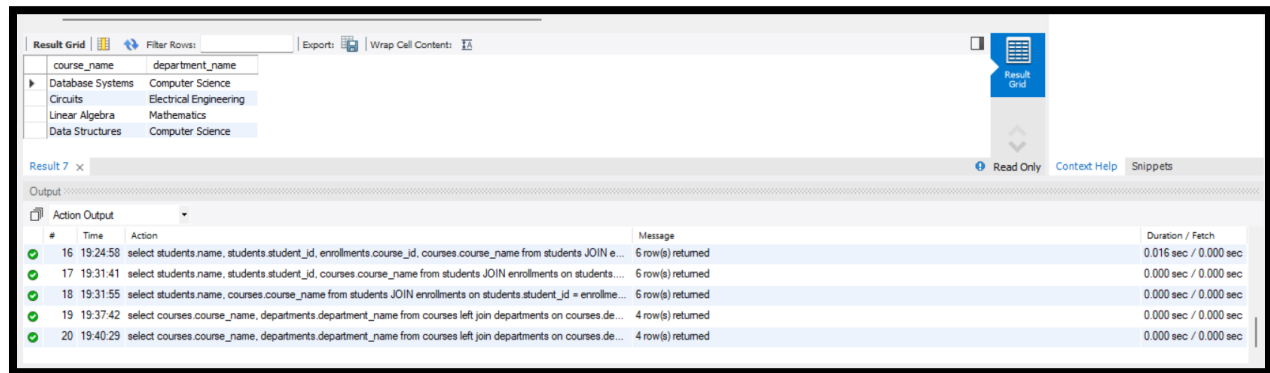


name	course_name
Alice	Database Systems
Bob	Circuits
Charlie	Database Systems
Diana	Linear Algebra
Alice	Data Structures
Charlie	Data Structures

#	Time	Action	Message	Duration / Fetch
14	19:14:28	select courses.course_name, Instructors.name from Courses left join Instructors on courses.instructor_id = Instructors.instructor_id	4 row(s) returned	0.000 sec / 0.000 sec
15	19:18:36	select courses.course_name, Instructors.name from Courses left join Instructors on courses.instructor_id = Instructors.instructor_id	4 row(s) returned	0.000 sec / 0.000 sec
16	19:24:58	select students.name, students.student_id, enrollments.course_id, courses.course_name from students JOIN enrollments on students.student_id = enrollments.student_id	6 row(s) returned	0.016 sec / 0.000 sec
17	19:31:41	select students.name, students.student_id, courses.course_name from students JOIN enrollments on students.student_id = enrollments.student_id JOIN courses on enrollments.course_id = courses.course_id	6 row(s) returned	0.000 sec / 0.000 sec
18	19:31:55	select students.name, courses.course_name from students JOIN enrollments on students.student_id = enrollments.student_id JOIN courses on enrollments.course_id = courses.course_id	6 row(s) returned	0.000 sec / 0.000 sec

Q3. Show course names and corresponding department names.

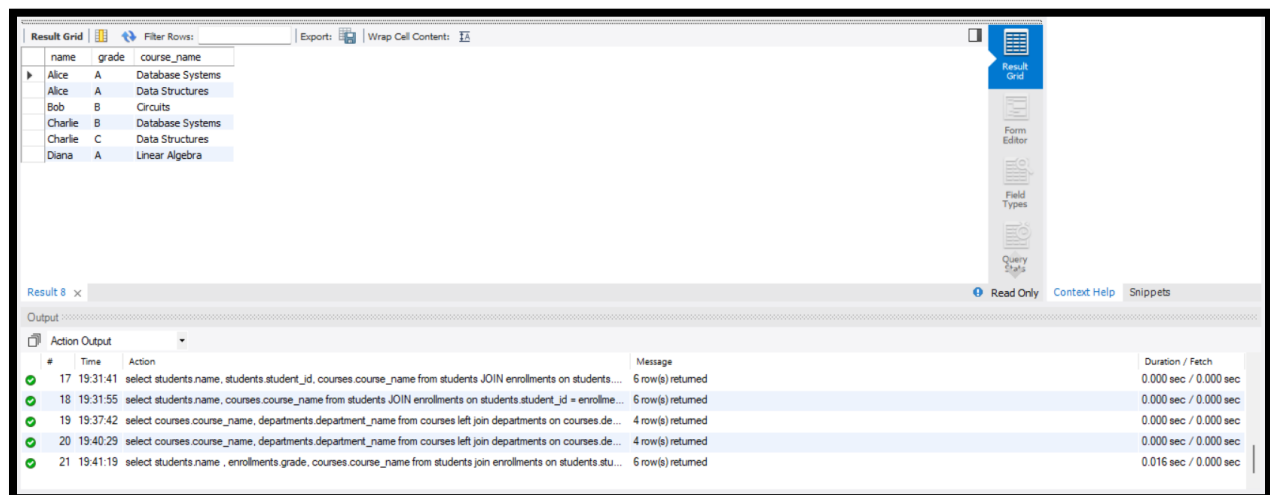
```
select courses.course_name, departments.department_name
from courses left join departments on courses.department_id = departments.department_id;
```



#	Time	Action	Message	Duration / Fetch
16	19:24:58	select students.name, students.student_id, enrollments.course_id, courses.course_name from students JOIN e...	6 row(s) returned	0.016 sec / 0.000 sec
17	19:31:41	select students.name, students.student_id, courses.course_name from students JOIN enrollments on students...	6 row(s) returned	0.000 sec / 0.000 sec
18	19:31:55	select students.name, courses.course_name from students JOIN enrollments on students.student_id = enrollme...	6 row(s) returned	0.000 sec / 0.000 sec
19	19:37:42	select courses.course_name, departments.department_name from courses left join departments on courses.de...	4 row(s) returned	0.000 sec / 0.000 sec
20	19:40:29	select courses.course_name, departments.department_name from courses left join departments on courses.de...	4 row(s) returned	0.000 sec / 0.000 sec

Q4. List all students with their grades and course names.

```
select students.name , enrollments.grade, courses.course_name
from students join enrollments on students.student_id = enrollments.student_id
join courses on courses.course_id = enrollments.course_id
order by students.name;
```



#	Time	Action	Message	Duration / Fetch
17	19:31:41	select students.name, students.student_id, courses.course_name from students JOIN enrollments on students...	6 row(s) returned	0.000 sec / 0.000 sec
18	19:31:55	select students.name, courses.course_name from students JOIN enrollments on students.student_id = enrollme...	6 row(s) returned	0.000 sec / 0.000 sec
19	19:37:42	select courses.course_name, departments.department_name from courses left join departments on courses.de...	4 row(s) returned	0.000 sec / 0.000 sec
20	19:40:29	select courses.course_name, departments.department_name from courses left join departments on courses.de...	4 row(s) returned	0.000 sec / 0.000 sec
21	19:41:19	select students.name , enrollments.grade, courses.course_name from students join enrollments on students.stu...	6 row(s) returned	0.016 sec / 0.000 sec

Q5. Show instructors and the courses they teach.

```
select Instructors.name , courses.course_name
```

```
from instructors join courses on instructors.instructor_id = courses.instructor_id
```

```
order by instructors.name;
```

The screenshot shows a database management tool interface. At the top, there's a 'Result Grid' tab with a table containing two columns: 'name' and 'course_name'. The data rows are:

name	course_name
Dr. Johnson	Circuits
Dr. Smith	Database Systems
Dr. Smith	Data Structures
Dr. Williams	Linear Algebra

Below the result grid, there's an 'Output' section with a dropdown menu set to 'Action Output'. It displays a log of database actions with columns: '#', 'Time', 'Action', 'Message', and 'Duration / Fetch'. The log shows several successful queries, including the one for Q5.

#	Time	Action	Message	Duration / Fetch
19	19:37:42	select courses.course_name, departments.department_name from courses left join departments on courses.de...	4 row(s) returned	0.000 sec / 0.000 sec
20	19:40:29	select courses.course_name, departments.department_name from courses left join departments on courses.de...	4 row(s) returned	0.000 sec / 0.000 sec
21	19:41:19	select students.name , enrollments.grade, courses.course_name from students join enrollments on students.stu...	6 row(s) returned	0.016 sec / 0.000 sec
22	19:42:28	select Instructors.name , courses.course_name from instructors join courses on instructors.instructor_id = cours...	4 row(s) returned	0.000 sec / 0.000 sec
23	19:45:00	select Instructors.name , courses.course_name from instructors join courses on instructors.instructor_id = cours...	4 row(s) returned	0.000 sec / 0.000 sec

Q6. Show the name of students who are enrolled in “Data Structures”.

```
select students.name
```

```
from students join enrollments on students.student_id = enrollments.student_id
```

```
join courses on enrollments.course_id = courses.course_id
```

```
where courses.course_name = "Data Structures"
```

```
order by students.name;
```

The screenshot shows a database management tool interface. At the top, there's a 'Result Grid' tab with a table containing one column: 'name'. The data rows are:

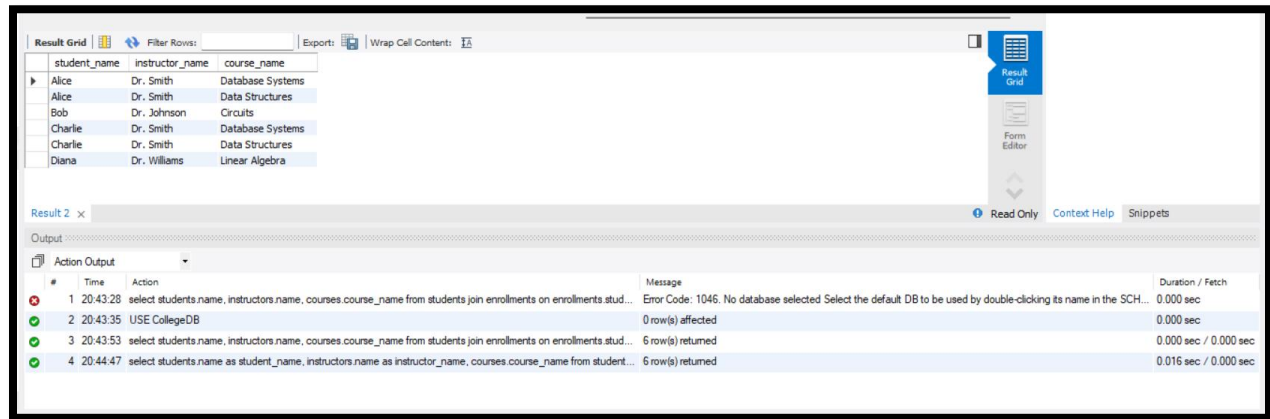
name
Alice
Charlie

Below the result grid, there's an 'Output' section with a dropdown menu set to 'Action Output'. It displays a log of database actions with columns: '#', 'Time', 'Action', 'Message', and 'Duration / Fetch'. The log shows several successful queries, including the one for Q6.

#	Time	Action	Message	Duration / Fetch
22	19:42:28	select Instructors.name , courses.course_name from instructors join courses on instructors.instructor_id = cours...	4 row(s) returned	0.000 sec / 0.000 sec
23	19:45:00	select Instructors.name , courses.course_name from instructors join courses on instructors.instructor_id = cours...	4 row(s) returned	0.000 sec / 0.000 sec
24	19:48:27	SELECT s.name FROM Students s JOIN Enrollments e ON s.student_id = e.student_id JOIN Courses c ON e.c...	2 row(s) returned	0.000 sec / 0.000 sec
25	19:48:37	select students.name, courses.course_name from students right join enrollments on students.student_id = enroll...	2 row(s) returned	0.000 sec / 0.000 sec
26	19:49:22	select students.name from students join enrollments on students.student_id = enrollments.student_id join cours...	2 row(s) returned	0.000 sec / 0.000 sec

Q7. List all students and their instructors for each course.

```
select students.name as student_name, instructors.name as instructor_name, courses.course_name
from students join enrollments on enrollments.student_id = students.student_id
join courses on enrollments.course_id = courses.course_id
join instructors on instructors.instructor_id = courses.instructor_id
order by students.name;
```



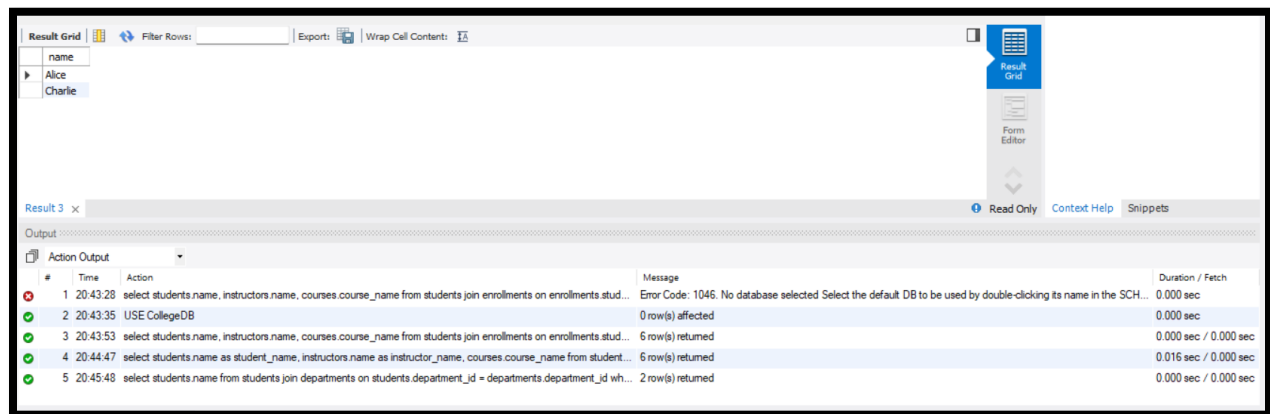
The screenshot shows a database IDE interface. At the top, there's a 'Result Grid' tab with a table containing student and instructor names for various courses. Below it, the 'Output' pane shows the 'Action Output' log for a series of SQL commands. The log includes a message about a missing database and the successful execution of a query that returned 6 rows.

student_name	instructor_name	course_name
Alice	Dr. Smith	Database Systems
Alice	Dr. Smith	Data Structures
Bob	Dr. Johnson	Circuits
Charlie	Dr. Smith	Database Systems
Charlie	Dr. Smith	Data Structures
Diana	Dr. Williams	Linear Algebra

#	Time	Action	Message	Duration / Fetch
1	20:43:28	select students.name, instructors.name, courses.course_name from students join enrollments on enrollments.student_id = students.student_id join courses on enrollments.course_id = courses.course_id join instructors on instructors.instructor_id = courses.instructor_id order by students.name;	Error Code: 1046. No database selected Select the default DB to be used by double-clicking its name in the SCH...	0.000 sec
2	20:43:35	USE CollegeDB	0 row(s) affected	0.000 sec
3	20:43:53	select students.name, instructors.name, courses.course_name from students join enrollments on enrollments.student_id = students.student_id join courses on enrollments.course_id = courses.course_id join instructors on instructors.instructor_id = courses.instructor_id order by students.name;	6 row(s) returned	0.000 sec / 0.000 sec
4	20:44:47	select students.name as student_name, instructors.name as instructor_name, courses.course_name from students join enrollments on enrollments.student_id = students.student_id join courses on enrollments.course_id = courses.course_id join instructors on instructors.instructor_id = courses.instructor_id order by student_name;	6 row(s) returned	0.016 sec / 0.000 sec

Q8. Show all students from the “Computer Science” department.

```
select students.name
from students join departments on students.department_id = departments.department_id
where departments.department_name = 'Computer Science';
```



The screenshot shows a database IDE interface. At the top, there's a 'Result Grid' tab with a table containing student names. Below it, the 'Output' pane shows the 'Action Output' log for a series of SQL commands. The log includes a message about a missing database and the successful execution of a query that returned 2 rows.

name
Alice
Charlie

#	Time	Action	Message	Duration / Fetch
1	20:43:28	select students.name, instructors.name, courses.course_name from students join enrollments on enrollments.student_id = students.student_id join courses on enrollments.course_id = courses.course_id join instructors on instructors.instructor_id = courses.instructor_id order by students.name;	Error Code: 1046. No database selected Select the default DB to be used by double-clicking its name in the SCH...	0.000 sec
2	20:43:35	USE CollegeDB	0 row(s) affected	0.000 sec
3	20:43:53	select students.name, instructors.name, courses.course_name from students join enrollments on enrollments.student_id = students.student_id join courses on enrollments.course_id = courses.course_id join instructors on instructors.instructor_id = courses.instructor_id order by students.name;	6 row(s) returned	0.000 sec / 0.000 sec
4	20:44:47	select students.name as student_name, instructors.name as instructor_name, courses.course_name from students join enrollments on enrollments.student_id = students.student_id join courses on enrollments.course_id = courses.course_id join instructors on instructors.instructor_id = courses.instructor_id order by student_name;	6 row(s) returned	0.016 sec / 0.000 sec
5	20:45:48	select students.name from students join departments on students.department_id = departments.department_id where departments.department_name = 'Computer Science';	2 row(s) returned	0.000 sec / 0.000 sec

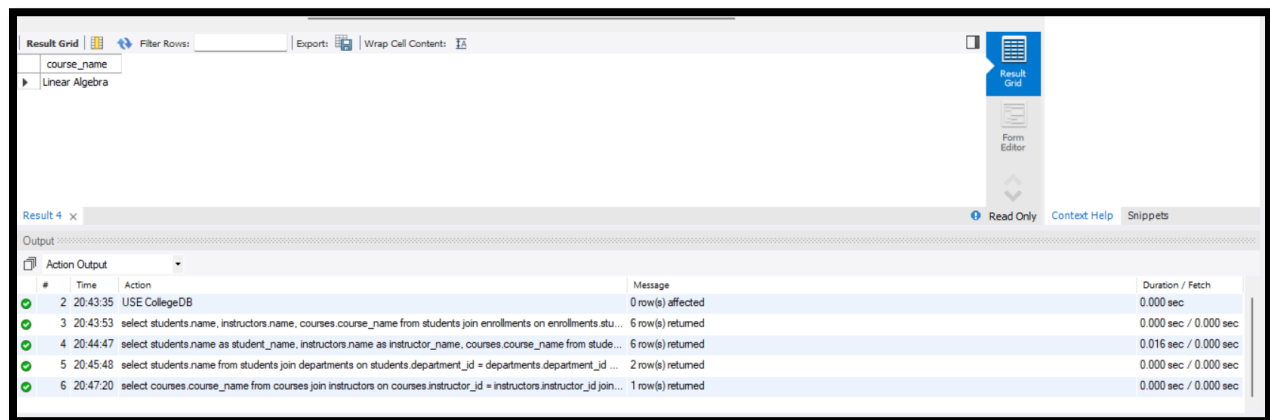
Q9. Show all courses taught by instructors from the “Mathematics” department.

```
select courses.course_name
```

```
from courses join instructors on courses.instructor_id = instructors.instructor_id
```

```
join departments on instructors.department_id = departments.department_id
```

```
where departments.department_name = 'Mathematics';
```



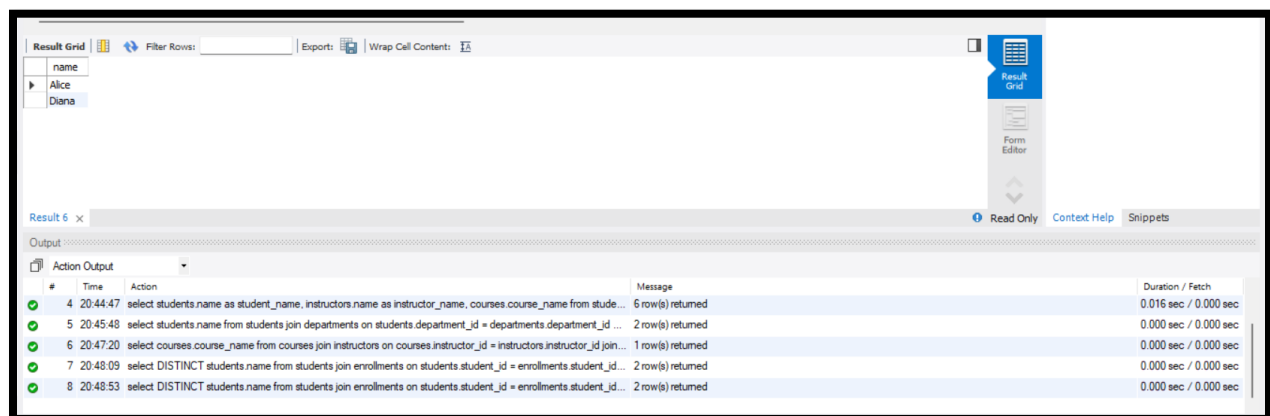
#	Time	Action	Message	Duration / Fetch
2	20:43:35	USE CollegeDB	0 row(s) affected	0.000 sec
3	20:43:53	select students.name, instructors.name, courses.course_name from students join enrollments on enrollments.stu...	6 row(s) returned	0.000 sec / 0.000 sec
4	20:44:47	select students.name as student_name, instructors.name as instructor_name, courses.course_name from stude...	6 row(s) returned	0.016 sec / 0.000 sec
5	20:45:48	select students.name from students join departments on students.department_id = departments.department_id ...	2 row(s) returned	0.000 sec / 0.000 sec
6	20:47:20	select courses.course_name from courses join instructors on courses.instructor_id = instructors.instructor_id join...	1 row(s) returned	0.000 sec / 0.000 sec

Q10. List students who have received an 'A' grade.

```
select DISTINCT students.name
```

```
from students join enrollments on students.student_id = enrollments.student_id
```

```
where enrollments.grade = 'A';
```



#	Time	Action	Message	Duration / Fetch
4	20:44:47	select students.name as student_name, instructors.name as instructor_name, courses.course_name from stude...	6 row(s) returned	0.016 sec / 0.000 sec
5	20:45:48	select students.name from students join departments on students.department_id = departments.department_id ...	2 row(s) returned	0.000 sec / 0.000 sec
6	20:47:20	select courses.course_name from courses join instructors on courses.instructor_id = instructors.instructor_id join...	1 row(s) returned	0.000 sec / 0.000 sec
7	20:48:09	select DISTINCT students.name from students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec
8	20:48:53	select DISTINCT students.name from students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec

Q11. Show students enrolled in courses taught by “Dr. Smith”.

```
select DISTINCT students.name
```

```
from Students join enrollments on students.student_id = enrollments.student_id
```

```
join Courses on enrollments.course_id = courses.course_id
```

```
join instructors on courses.instructor_id = instructors.instructor_id
```

```
WHERE instructors.name = 'Dr. Smith';
```

name
Alice
Charlie

#	Time	Action	Message	Duration / Fetch
7	20:48:09	select DISTINCT students.name from students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec
8	20:48:53	select DISTINCT students.name from students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec
9	20:50:25	select DISTINCT students.name from students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec
10	20:51:44	select DISTINCT students.name from Students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec
11	20:51:54	select DISTINCT students.name from Students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec

Q12. Display course names along with the number of students enrolled.

```
SELECT courses.course_name, count(enrollments.student_id) AS total_students
```

```
FROM courses
```

```
LEFT JOIN enrollments ON courses.course_id = enrollments.course_id
```

```
GROUP BY courses.course_id, courses.course_name;
```

course_name	total_students
Database Systems	2
Circuits	1
Linear Algebra	1
Data Structures	2

#	Time	Action	Message	Duration / Fetch
8	20:48:53	select DISTINCT students.name from students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec
9	20:50:25	select DISTINCT students.name from students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec
10	20:51:44	select DISTINCT students.name from Students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec
11	20:51:54	select DISTINCT students.name from Students join enrollments on students.student_id = enrollments.student_id...	2 row(s) returned	0.000 sec / 0.000 sec
12	20:53:42	select courses.course_name, count(enrollments.student_id) as total_students from courses left join enrollments ...	4 row(s) returned	0.016 sec / 0.000 sec

Q13. List departments and the number of students in each.

```
select departments.department_name , count(students.student_id) as total_students
from departments
left join students on departments.department_id = students.department_id
group by departments.department_id, departments.department_name;
```

department_name	total_students
Computer Science	2
Electrical Engineering	1
Mathematics	1

#	Time	Action	Message	Duration / Fetch
10	20:51:44	select DISTINCT students name from Students join enrollments on students.student_id = enrollments.student_id	2 row(s) returned	0.000 sec / 0.000 sec
11	20:51:54	select DISTINCT students name from Students join enrollments on students.student_id = enrollments.student_id	2 row(s) returned	0.000 sec / 0.000 sec
12	20:53:42	select courses.course_name, count(enrollments.student_id) as total_students from courses left join enrollments	4 row(s) returned	0.016 sec / 0.000 sec
13	20:55:25	select departments.department_name , count(students.id) as total_students from departments left join students	Error Code: 1054. Unknown column 'students.id' in field list	0.016 sec
14	20:55:37	select departments.department_name , count(students.student_id) as total_students from departments left join s	3 row(s) returned	0.000 sec / 0.000 sec

Q14. List courses that have more than one student enrolled.

```
select courses.course_name, count(enrollments.student_id) as 'No.of Students Enrolled'
from Courses
join Enrollments on courses.course_id = enrollments.course_id
group by courses.course_id, courses.course_name
having count(enrollments.student_id) > 1;
```

course_name	No.of Students Enrolled
Database Systems	2
Data Structures	2

#	Time	Action	Message	Duration / Fetch
11	20:51:54	select DISTINCT students name from Students join enrollments on students.student_id = enrollments.student_id	2 row(s) returned	0.000 sec / 0.000 sec
12	20:53:42	select courses.course_name, count(enrollments.student_id) as total_students from courses left join enrollments	4 row(s) returned	0.016 sec / 0.000 sec
13	20:55:25	select departments.department_name , count(students.id) as total_students from departments left join students	Error Code: 1054. Unknown column 'students.id' in field list	0.016 sec
14	20:55:37	select departments.department_name , count(students.student_id) as total_students from departments left join s	3 row(s) returned	0.000 sec / 0.000 sec
15	20:57:33	select courses.course_name, count(enrollments.student_id) as 'No.of Students Enrolled' from Courses join Enroll	2 row(s) returned	0.000 sec / 0.000 sec

Q15. Show names of students not enrolled in any course.

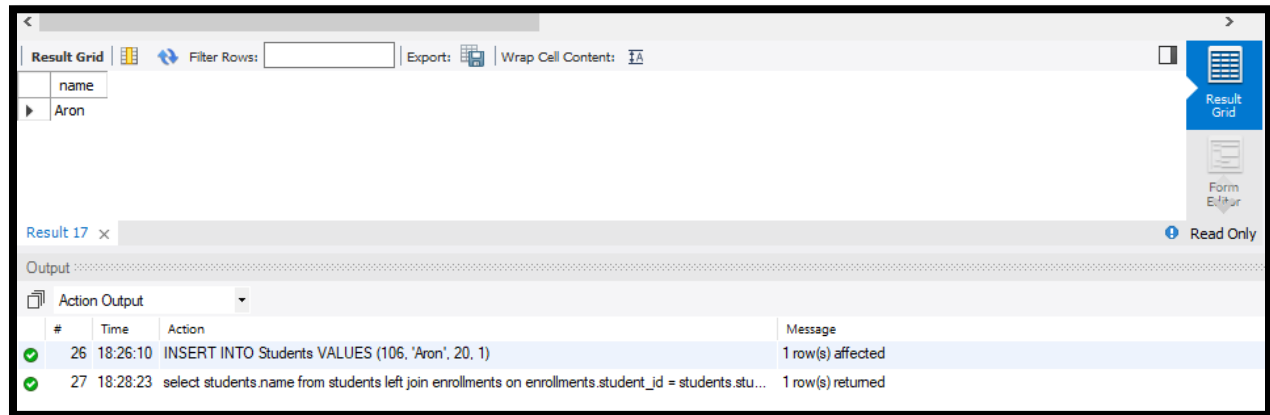
```
INSERT INTO Students VALUES
```

```
(106, 'Aron', 20, 1);
```

```
SELECT students.name FROM students
```

```
LEFT JOIN enrollments ON enrollments.student_id = students.student_id
```

```
WHERE enrollments.student_id is NULL;
```



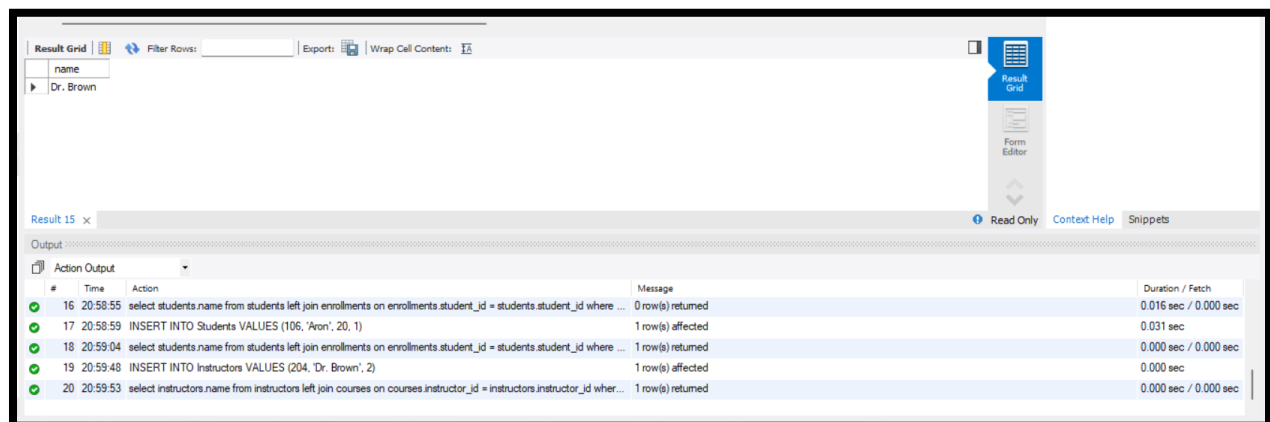
Q16. Show instructors who are not teaching any course.

```
INSERT INTO Instructors VALUES (204, 'Dr. Brown', 2);
```

```
select instructors.name
```

```
from instructors left join courses on courses.instructor_id = instructors.instructor_id
```

```
where courses.instructor_id is NULL ;
```



Q17. Show all students and all courses, even if the student is not enrolled (FULL OUTER JOIN emulation).

```
select students.name , courses.course_name
```

```
from students left join enrollments on enrollments.student_id = Students.student_id
```

```
left join courses on courses.course_id = enrollments.course_id
```

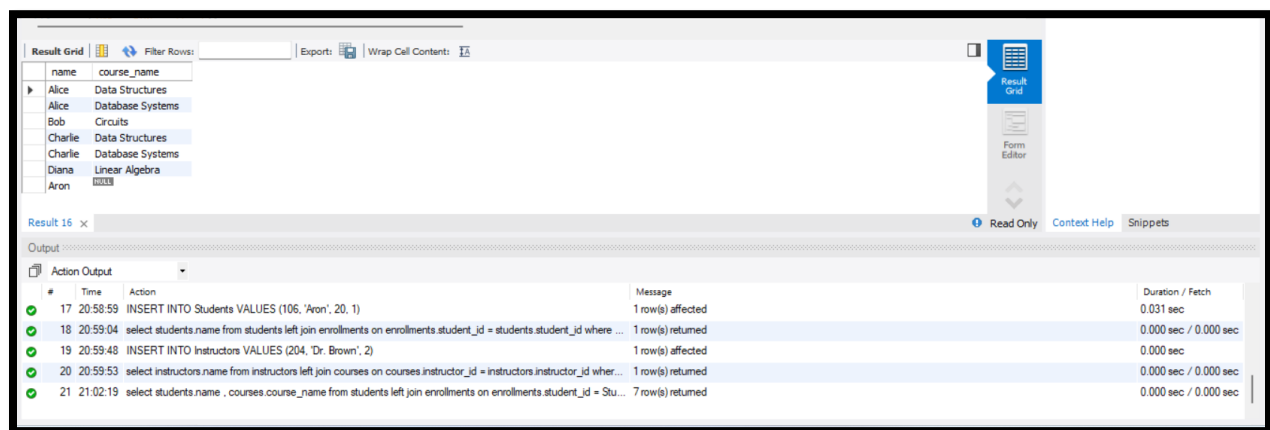
union

```
select students.name, courses.course_name
```

```
from students
```

```
right join enrollments on enrollments.student_id = students.student_id
```

```
right join courses on enrollments.course_id = courses.course_id;
```



Q18. List all students enrolled in courses from departments other than their own.

```
INSERT INTO Students VALUES
```

```
(105, 'Ben', 21, 1);
```

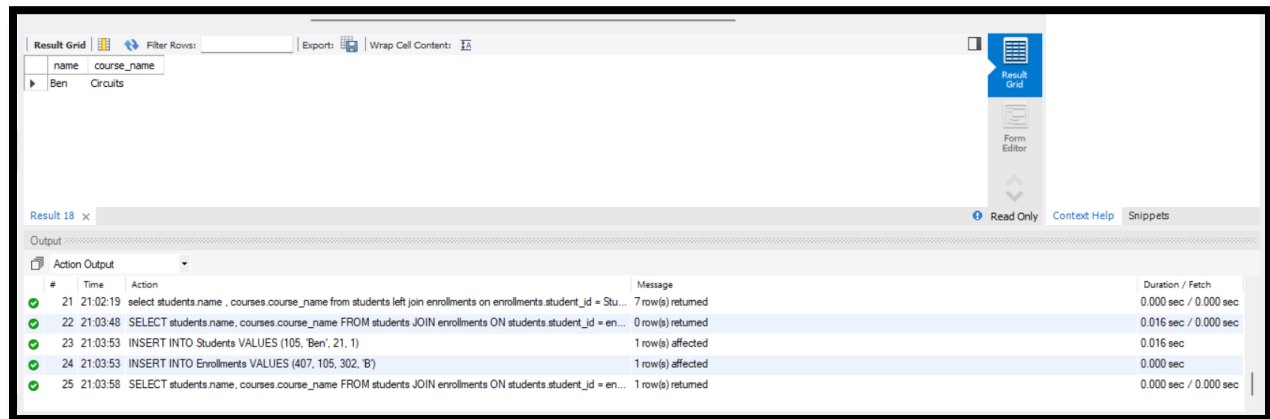
```
INSERT INTO Enrollments VALUES
```

```
(407, 105, 302, 'B');
```

```

SELECT students.name, courses.course_name FROM students
JOIN enrollments ON students.student_id = enrollments.student_id
JOIN courses ON enrollments.course_id = courses.course_id
WHERE students.department_id <> courses.department_id;

```



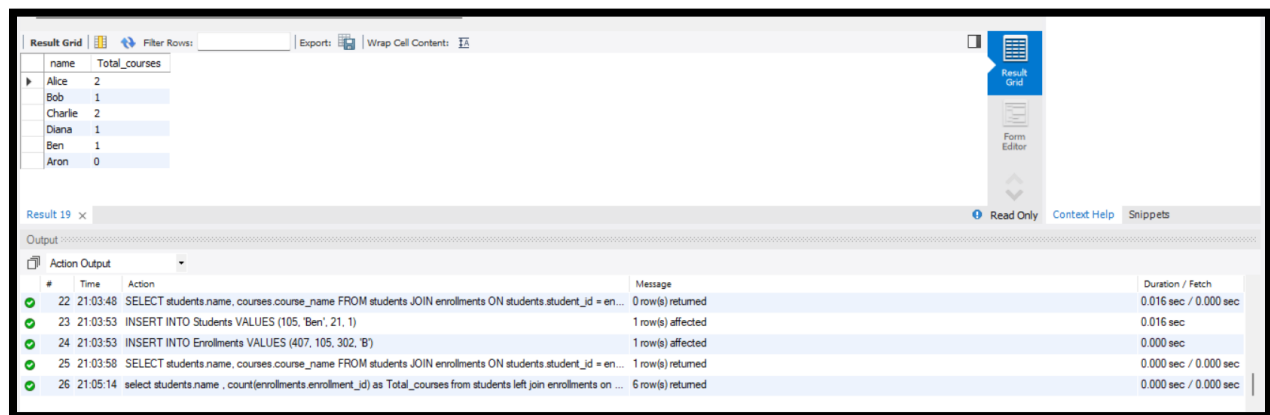
#	Time	Action	Message	Duration / Fetch
21	21:02:19	select students.name , courses.course_name from students left join enrollments on enrollments.student_id = Stu...	7 row(s) returned	0.000 sec / 0.000 sec
22	21:03:48	SELECT students.name, courses.course_name FROM students JOIN enrollments ON students.student_id = en...	0 row(s) returned	0.016 sec / 0.000 sec
23	21:03:53	INSERT INTO Students VALUES (105, 'Ben', 21, 1)	1 row(s) affected	0.016 sec
24	21:03:53	INSERT INTO Enrollments VALUES (407, 105, 302, 'B')	1 row(s) affected	0.000 sec
25	21:03:58	SELECT students.name, courses.course_name FROM students JOIN enrollments ON students.student_id = en...	1 row(s) returned	0.000 sec / 0.000 sec

Q19. List all students and the total number of courses they are enrolled in.

```

select students.name , count(enrollments.enrollment_id) as Total_courses
from students left join enrollments on students.student_id = enrollments.student_id
group by students.student_id, students.name;

```



#	Time	Action	Message	Duration / Fetch
22	21:03:48	SELECT students.name, courses.course_name FROM students JOIN enrollments ON students.student_id = en...	0 row(s) returned	0.016 sec / 0.000 sec
23	21:03:53	INSERT INTO Students VALUES (105, 'Ben', 21, 1)	1 row(s) affected	0.016 sec
24	21:03:53	INSERT INTO Enrollments VALUES (407, 105, 302, 'B')	1 row(s) affected	0.000 sec
25	21:03:58	SELECT students.name, courses.course_name FROM students JOIN enrollments ON students.student_id = en...	1 row(s) returned	0.000 sec / 0.000 sec
26	21:05:14	select students.name , count(enrollments.enrollment_id) as Total_courses from students left join enrollments on ...	6 row(s) returned	0.000 sec / 0.000 sec